MINIMUM PROFESSIONAL QUALIFICATIONS (Duties, Responsibilities, Experience and Education Requirements).

Principal
This person(s) shall have a Master's Degree or higher in Biology, Environmental Science, Natural Science, or closely related field as required by the deliverables of Section C, and approved by the Contracting Officer. This person(s) shall also have a minimum of 10 years full-time demonstrated professional experience applying theories, methods, and practices of biology with knowledge in the Asia-Pacific region that enables them to make professional judgment about the identification, evaluation, and documentations of biological resources. The person must have knowledge of natural resource management, ecology, and laws to protect the natural environment. This person(s) shall be able to provide oversight, guidance, quality control; and be responsible for ensuring the professional quality and technical accuracy of all field work, report writing, laboratory analyses, graphics, and geographic information systems (GIS) conducted under this contract. This person shall verify the accuracy and integrity of all efforts by executing the signature page of each deliverable, and shall be responsible to make sure the electronic product meets the scope of work and is compatible with the Government systems.

Project Manager (Biological)
Person(s) filling this position shall have a Master's Degree or 5 years working experience in Natural Sciences such as Biology, Botany, or Environmental Science, as required by the deliverable of Section C, or closely related field, approved by the Contracting Officer. The person(s) shall also have 4 years full-time demonstrated professional experience working as project manager for natural resources studies, surveys, organizing projects, and environmental media areas in the Asia-Pacific region that enables professional judgments to be made. This person(s) must have demonstrated ability to manage and implement research design and sampling strategies; make strategic and professional decision in the field; and have oral and written communication skills as evidenced by their completed technical reports/publications. This person(s) shall provide direction in the field, execution of document development; and shall oversee all field or laboratory work, geographic information systems (GIS), statistical analysis, and shall analyze and interpret data, and present synthesis of findings in a written report and/or GIS, as required by the task order.

Environmental Scientist
Person(s) filling this position shall have a Bachelor's Degree or higher in Biology or related field as required by the deliverable in Section C, including Environmental Science, Botany, Ecology, or other natural sciences. The person(s) shall also have 2 years full-time demonstrated professional experience working with the environmental media listed in Section C. This person(s) must have knowledge and experience working with applicable U.S. regulations and experience in developing environmental plans and studies. Duties include analyzing and interpreting data, writing plans and studies, as required by the task order.

Technical Biologist
At minimum this person(s) shall have a degree in Biology, Environmental Science, Botany, Ecology, or related wildlife or biological field. A minimum experience of two (2) years in planning and conducting wildlife field inventories and monitoring studies, interpreting data,

preparing wildlife management plans, survey reports, and recommending habitat management practices. This person must have demonstrated abilities in planning assessment surveys, follow scientific and standard survey methods, executing work in a field environment, hike across difficult terrain, and in adverse weather conditions and potentially remote locations.

NEPA Planner
At a minimum this person shall have a Bachelor's degree or Master's degree in environmental planning, environmental science, biology or related field. Have a Minimum 5 years' professional experience in NEPA compliance and permitting. Have demonstrated experience managing Environmental Impact Statements (EISs), Environmental Assessments (EAs), or Record of Categorical Exclusions (RCEs). Have experience working with regulatory agencies and the public. This person(s) would serve as the NEPA document writer that would prepare NEPA related compliance documents, legal and regulatory documents, and technical reports/deliverables. Duties would include environmental resource consulting and project management, especially in the preparation of federal, state, and overseas environmental guidelines.

GIS Specialist
If the GIS Specialist is not also the Project Manager, Environmental Scientist, or Technical Biologist, this person must have a Bachelor's Degree or higher in geographic information systems, geography, natural resources, natural sciences, planning, engineering, or related field. This person shall be responsible to make sure the electronic product meets the scope of work and is compatible with the Government systems described in Attachment J-4. This person shall be able to demonstrate experience in Geographic Information System analysis, data organization, geodatabase maintenance and troubleshooting, and geospatial analyses; ability to utilize and import and maximize the accuracy of GIS data from a variety of GPS data sources (Terrasync, smart phone maps, Garmin, Trimble, etc.); ability to work with ERDAS, AutoCAD, or other locational information software and an ability to transfer files between different locational information software types. Ability to collect GPS data in a variety of settings including forests, steep slopes, adverse weather conditions and remote locations. They shall have certified training and professional experience in creating and maintaining geographic information systems that are compatible with Arc View 8 or later versions and using AutoCAD and Microsoft Access databases.

~~**Entomologist**~~
~~Person(s) filling this position shall have a Master's Degree in entomology or 5 years of experience working in entomology or pest management. Have demonstrated ability to identify pests, implement and organize HACCP plans, implement extermination plans, recommend pesticides at recommended label quantities, conduct pest assessments, trap insects, and related subjects required in Section C. Knowledgeable in the ecology of pest species, invasive species, and incursion pathways. Person(s) shall have State or Federal certifications to apply EPA herbicides/pesticides. Have demonstrated experience educating DoD members, civilians and the public about pests and invasive species.~~

Marine Biologist
The Marine Biologist shall provide technical oversight and technical quality control of marine resources work performed. Together with the Project Manager, the Marine Biologist shall ensure that associated

deliverables meet or exceed professional standards. In addition to working closely with the Project Manager, the Marine Biologist shall coordinate closely with the Government's Marine Resources SMEs. The Marine Biologist must have the following qualifications:
a. Possess a bachelor's degree or higher in a marine science-based field, such as biology, ecology, environmental science or other closely related field that is approved by the Contracting Officer.
b. Shall have at least five years of full-time professional experience in a marine science-based field, such as biology, ecology, environmental science, or closely related field. Of the five years of working experience in the marine sciences, at least two years shall have been in support of the development of reports used in NEPA documents including, but not limited to Endangered Species Act consultations, Marine Mammal Protection Act Consultations, Essential Fish Habitat Assessments.
c. Shall have four years of full-time demonstrated professional experience working as project manager for marine resources studies, conducting marine resources surveys, organizing projects, and coordinating with other environmental media areas. This person must have demonstrated ability to successfully manage and implement research design and sampling strategies; make strategic and professional decisions in the field; and clearly communicate both orally and in writing as evidenced by their completed technical reports/publications.
d. Shall have experience including, but not limited to, conducting underwater surveys for coral, fish, and other species in Hawaii, Japan, and other Pacific islands, acoustic monitoring and analysis, etc.
e. Shall have a diver certification (e.g., America Academy of Underwater Sciences [AAUS], commercial) and provide current active status and must be familiar with and follow the US Army Corps of Engineers EM 385-1-1, Safety and Health Requirements Manual.
f. Shall have the current ability and qualifications to prepare, lead, and conduct the on-site marine resource surveys described in this SOW on Tinian.
g. May be an employee of the prime firm or a subcontractor.

Hydrologist/ Wetland Specialist

Person(s) filling this position must have bachelor's degree in hydrology or related natural sciences; knowledgeable in the ramifications of requesting a Preliminary Jurisdictional Delineation (PJD) or approved JD (AJD) with Army Corp of Engineers. This person shall have extensive experience identifying wetland and non-wetland waters of the U.S. and Pacific Islands of both the 1987 Army Corps of Engineering wetland delineation manual or later versions, and the regional supplements appropriate to the Task Order area, and knowledgeable dealing with the delineation process. The person(s) must have experience getting delineations approved by the ACOE, and ability to correctly identify to indicator species, subspecies or variety (as appropriate) all vascular plant species that occur on project site as defined in Task Order using taxonomic keys if necessary. This person shall have formal documented training in the wetland delineation methods used in the US Army Corps of Engineers Wetland Delineation Manual (Environmental Laboratory, 1987) and regional supplements.

~~**Arborist**~~

~~This person(s) should be a Certified Arborist by the International Society of Arboriculture and have over three (3) years documented experience. They shall assess the health and safety of trees and execute corrective actions. Duties include inspections and diagnoses of trees for pests or indications of pest injury or infestation, and the diagnosis and treatment of tree structure support systems. The person may be required to climb trees and comply with industry standards~~

~~established by ANSI Z133.1-1994 and the International Society of Arboriculture for urban forestry climbing and with the use of aerial lift equipment and all other equipment required in the performance of the job. Plan strategies and execute plans to stabilize the structure of trees to prevent it from toppling, and remove dead or diseased parts of the tree to improved structure and tree health.~~

Site Safety Officer
The SSHO, at a minimum, must have completed the 30-hour OSHA Construction safety class or have 30 hours of verifiable formal construction safety and health training covering the subjects of the OSHA 30-hour course applicable to the work to be performed and given by qualified instructors. The SSHO is also required to have five (5) years of construction industry safety experience or three (3) years of experience and possessing a Certified Safety Professional (CSP) or safety and health degree. An SSHO or a Designated Representative, as identified in the APP/AHA and as deemed appropriate/equivalent to SSHO by the GDA, shall be on-site at all times when work is being performed. The SSHO shall be responsible for managing, implementing and enforcing the Contractor's Safety and Health Program in accordance with the accepted APP. SSHOs shall maintain this competency through 24 hours of formal safety and health related coursework every four (4) years. For limited service contracts, (e.g. visual inspections) the Contracting Officer and Safety Office may waive the more stringent SSHP elements of this section. For complex or high hazard projects, the SSHO shall have a minimum of ten (10) years of safety-related work with at least five (5) years of experience on similar type projects.

Editor
This person must have a bachelor's degree in English or related field and three (3) years of professional editorial experience with products and activities that demonstrate excellent writing, grammatical, and editing skills and ability to edit authors' original manuscripts, including rewriting as necessary. Background in biology, natural resources or an environmental field is desirable.

Administrative Assistant Support
This person must have a high school diploma or equivalent and 1 year of general experience, or 2 years of education above the high school level. Duties are to perform work of a general clerical or administrative nature for which computer skills and knowledge of such things as spreadsheets, word processing, data entry, and invoicing are required.